Partha Sarathi Purkayastha

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Research Profile

I build and evaluate AI systems that adapt to people and remain reliable in consequential settings. My research spans verification for LLM tutoring, dynamic student simulation, context-aware computer vision, and multimodal AI for healthcare.

Education

2025 – 2027 **M.Sc. Computer Science**, ETH Zürich, Zürich, Switzerland
Relevant Coursework: Probabilistic Artificial Intelligence, Large Language Models, Mixed Reality, AI in Education, Big Data, Information Security Lab, Computational Intelligence Lab, Computational Statistics

2020 – 2024 **B.E. Computer Science**, Birla Institute of Technology and Science, Pilani, India
CGPA: 9.83/10, **Director's Gold Medal Awardee**
Relevant Coursework: Data Structures and Algorithms, Object Oriented Programming, Database Systems, Theory of Computation, Operating Systems, Computer Networks, Linear Algebra, Probability and Statistics, Discrete Mathematics, Artificial Intelligence, Computer Graphics

Publications

Journal & Conference Papers

AIED 2026 **Partha Sarathi Purkayastha**. Reliability as a Teammate: Budgeted Verifier-in-the-Loop (BVIL) Policies for Reliable LLM Tutoring Actions. *Published in Artificial Intelligence in Education (AIED 2026), Springer LNAI 16583.* [Paper]

IJCV 2026 Tajamul Ashraf, Rajes Manna*, **Partha Sarathi Purkayastha***, Tavaheed Tariq, Janibul Bashir. Context Aware Grounded Teacher for Source Free Object Detection. *International Journal of Computer Vision (IJCV), 134, Article 306 (2026).* *Equal contribution. [Paper]

Workshop Papers

EMNLP-W 2026 **Partha Sarathi Purkayastha**, Heejin Do. DySAS: Dynamic State-Aware Student Simulation Beyond Static Personas. *REALM Workshop at EMNLP 2026; published in the ACL Anthology.* [Paper]

ICML-W 2026 **Partha Sarathi Purkayastha**. When Thinking Before Retrieval Hurts: TraceBound Diagnostics for Adaptive Knowledge-Graph Retrieval. *Accepted at the Failure Modes in Agentic AI (FAGEN) Workshop at ICML 2026.* [Paper]

Preprints

2026 Nachiket Kotalwar, Conrad Borchers, Shubham Gandhi, **Partha Sarathi Purkayastha**, David Yaron, Carolyn P. Rosé. Exploring Visual Access and Proactive AI Tutoring for Handwritten Mathematics. *arXiv:2610.26411, 2026.* [Paper]

2026 Tajamul Ashraf, Rozain Shakeel, Tavaheed Tariq, **Partha Sarathi Purkayastha**, Moloud Abdar, Janibul Bashir, Fida Mohammad Thoker, Adams Wai Kin Kong. From Models to Agents: A Closer Look into Agentic AI Systems. *Preprints.org, 2026.* [Paper]

Awards

2026 **ASIA Data Challenge, Track 2 Winner: Longitudinal ISNCSCI Imputation**.
Won the American Spinal Injury Association challenge for reconstructing full neurological assessments from expedited exams. Presented the winning work at the ASIA Annual Scientific Meeting in San Antonio, April 2026, with a travel grant.

2020 – 2024 **Institute Merit Scholarship, BITS Pilani**: Full tuition waiver awarded from the start of the B.E. for consistently ranking in the top 1%.

Research Experience

May'26 – Present **Research Assistant**, ETH SCAI Lab, Zürich / Nottwil, Switzerland.
- Analyzed wrist-signal correlations with **BNP (cardiac marker)**, weight and fluid balance for **Decongestion Monitoring**.
- Built a shared encoder on 11 **wearable activity** datasets; adapted to 5 cohorts across varied sensors, conditions and tasks.
- Analyzed **PPG** signals across modalities and encoders for a **cuffless blood pressure** transfer strategy (finger-to-wrist).

Supervisor: **Dr. Diego Paez-Granados**

Sept'25 – Sept'26 **Student Researcher**, ETH LRE Lab, Zürich, Switzerland.
- Developed a **psychologically grounded, dynamic state-aware framework** for simulating **diverse and evolving student behaviours** beyond static personas, enabling rigorous evaluation of **adaptive LLM tutors** across multi-turn interactions. *Published in the REALM @ EMNLP 2026 proceedings.*

Supervisors: **Dr.Heejin Do**, **Prof.Mrinmaya Sachan**

Jul'24 – Jun'25 **Research Fellow**, Microsoft Research India, Bangalore, India.
○ Developed a **privacy taxonomy** and curated a **context-dependent summarization dataset**; evaluated **LLMs** using automatic metrics and human judgments, outperforming **GPT-4o** on privacy adherence while maintaining summary **consistency, relevance, and coherence**.

Supervisors: **Dr. Akshay Nambi**, **Dr. Venkat Padmanabhan**

Aug'23 – Jan'24 **Research Intern**, BITS Pilani, Pilani Campus, India.
○ Built a **fundus-image annotation tool** and collaborated with a hospital to curate a **3,000-image dataset**; applied image processing and custom deep-learning models to automate lesion annotation, achieving **over 80% accuracy and specificity**.

Supervisor: **Dr.Sundaresan Raman**

Jan'23 – May'23 **Research Intern**, Mercedes-Benz, Bangalore, India.
○ Enhanced component design by training **customized VAE models** for **parameter-controlled generation** on a **25,000+ point proprietary dataset**, and deploying an **application-style website** enabling user-friendly custom design generation for company research.

Supervisor: **Dr.Poonam Goyal**

Industry Experience

Jan'24 – Jul'24 **SWE Intern**, Sprinklr, Gurgaon, India.
○ Identified and resolved **memory leaks** using **memlab**, reducing memory usage by **10%**. Implemented **server-side rendering** with **Next.js** and **GraphQL**, improving load times by **20%**; removed render blockers to improve application stability.

May'23 – Aug'23 **SWE Intern**, Uber, Hyderabad, India.
○ Developed an **end-to-end application** using **React.js**, **GraphQL**, and **GoLang** to manage **credit types** for **2000+ employees**. Migrated financial data to a **MySQL**-based centralized database for **secure transactions**. Automated integration and end-to-end testing, boosting system reliability and cutting onboarding time by **two months**.

Teaching Experience

Sept'25 – Jun'26 **AI and Digital Technology**, ETH Zürich, Teaching Assistant. **Supervisor:** **Dr. Clara Meister**.
Jan'23 – Dec'23 **Techniques in Social Research**, BITS Pilani, Teaching Assistant. **Supervisor:** **Dr. Tanu Shukla**.
Jan'23 – Dec'23 **Introductory Psychology**, BITS Pilani, Teaching Assistant. **Supervisor:** **Dr. Tanu Shukla**.
Jan'22 – May'22 **Microprocessors and Interfacing**, BITS Pilani, Teaching Assistant. **Supervisor:** **Dr. Vinay Chamola**.

Academic Service

2026 **Program Committee Member**: AI4GOOD and FAGEN (ICML-W 2026); REALM (EMNLP-W 2026).

Community Activities

Oct'24 – Nov'24 **Volunteer**, *Vision Empowerment*, Microsoft Research India, Bangalore, India. Designed **Braille-based aids** to help **vision-impaired** children in basic **mathematical** learning through **haptic feedback**.

Jan'22 – Dec'22 **Publicity Coordinator**, *Arunodoi*, BITS Pilani, India. Led the **Media-editing** team for **Arunodoi**, promoting **Northeast India**'s cultural awareness and community engagement.

Nov'21 – Jan'22 **Volunteer**, *Indian Red Cross Society*, BITS Pilani, India. Supported a **two-day long** blood donation camp with **3000+** footfall and **800+** donations.

Additional Projects

Jan'26 – Apr'26 **ASIA Data Challenge: Neurological Assessment Reconstruction**.
○ Built the team's **leakage-safe, target-wise TabPFN pipeline** for reconstructing missing sensory scores, preserving observed follow-up values and applying conservative anatomical corrections for domain-specific *contextualization*. [Code]

Supervisor: **Dr. Diego Paez-Granados**

Sept'25 – Jan'26 **NeuroReach VR: Mixed-Reality Rehabilitation System for Stroke Recovery**.
○ Built a pass-through mixed-reality rehabilitation system on **Meta Quest 3S** with **Logitech MX Ink** for gross- and fine-motor training, clinician-readable event-level CSV logging, and a **10-user pilot study** achieving a **79.7/100** score on the **System Usability Scale (SUS)**.

Supervisors: **Marwen Mokni**, **Prof. Christoph Bauer**

Skills

Programming Python, Java, TypeScript/JavaScript, C/C++, Go, SQL, C#
ML & Data PyTorch, Transformers, TabPFN, NumPy, Pandas, Plotly, OpenCV
Development Node.js, React.js/Next.js, GraphQL, Tailwind CSS, Unity
Tools LaTeX, Git/GitHub, Docker, MemLab, Postman, MySQL, MongoDB